COMP 581: Introduction to Robotics

Fall 2026
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Lab 2: Wall Following

Deadline

October 27, 2026, at the beginning of class. For the late policy, see the course syllabus. **Before the due time, each team should submit their team's code in Gradescope**; see specific instructions in the Code section below**.**

Submission Deadlines:

Optional Pre-Lab Reflection:	10/26 at 11:59 pm
Submit Code:	10/27 at 9:30 am
Optional Post-Lab Reflection:	10/31 at 11:59 pm

For all submissions, include the **names** and **PIDs** for each team member and include the **team number**. Add all team members to the submission using the **View or edit group** option in Gradescope.

The Objective

The primary purpose of this lab is to implement forward kinematics and wall following for a mobile robot.

Robot Design Requirements

Your robot must be built only using pieces from your LEGO Mindstorms kit. You should design your robot such that the dark gray center button may be pressed without moving the robot and such that pushing the button does not disrupt any of the sensors. Your robot should remain in one piece at all times, its diameter with respect to the ground should not exceed 40 cm, and its diameter shall not change substantially during the task. All measurements will be made with respect to the ***measuring point*****:** a distinct point of your choosing that is on the **frontmost surface** of your robot. More specifically, you will place your robot on the ground behind a starting line drawn on the ground, such that your chosen measuring point is **directly above a point on the starting line that will be marked**, and no part of your robot is permitted to be in front of the starting line. Your robot should be designed to accomplish the task autonomously without any human intervention (except for pushing the dark gray center button as required).

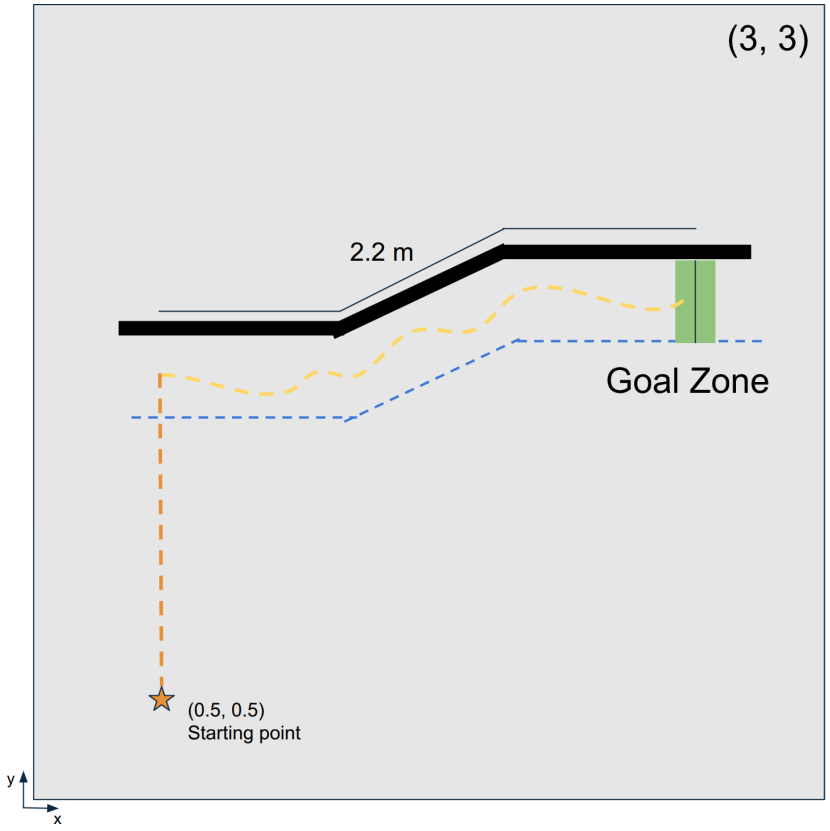
The Task

The task in this lab is to be completed in a single run, without any intervention.

Objective 1 (Detect wall): You will place your robot behind a starting line such that its **measuring point will be on a specific *starting point (marked)*** on the starting line (see figure below). You will then push the dark gray center button. Your robot should then move straight forward (in the positive *y* direction on the ground). Straight ahead, at a distance of somewhere between 75 and 130 cm, will be a wall that is perpendicular to the robot's forward motion. The wall will be at least 17 cm high, of unknown width, and it is bumpable and can be detected via ultrasound.

Objective 2 (Turn at wall): Once your robot is less than 30 centimeters from the wall, your robot should turn right and follow the wall.

Objective 3 (Following the wall): Once starting to follow the wall, your robot's measuring point should never be more than 30 cm from the wall at all times while following the wall.
The wall may be curved and may bend up to 45 degrees. The robot should follow the wall for a distance of 2.2 m, as measured by the distance of the wall (see figure), and then stop. It should then beep to announce that it has reached its destination.



All of the robot motion must be completed within a 3-meter square workspace without leaving the workspace. The starting point is at coordinate (0.5m, 0.5m). The task (from the time the button is pushed until the robot stops moving) must be completed in under 90 seconds.

In the figure above, we illustrate an example of a possible environment that complies with the rules. We note that the environment in your actual run may be different. The robot starts at the starting point. The wall is shown in black. The angles of any bend are unknown in advance, but they are at most 45 degrees with respect to the previous wall. *Note that there may be more than one bend*. The dotted blue line represents a 30 cm distance to the wall. The distance is measured along the walls from the point where the robot would ideally reach the wall. The goal zone is perpendicular to the wall. It is 10 cm long and 30 cm wide, i.e. if your robot stays within 30 cm from the wall and travels along the walls for 2.20m ± 5cm, you will receive full credit. The yellow dashed line shows a possible robot trajectory that would receive full credit if completed in the allotted time and assuming the robot stops inside the green goal zone.

Points:

Logistics (10 points):
Turn in code on time: 10 points.
Performance (90 points):

- **Attempt**: 30 points. You will receive these points simply by having EV3 MicroPython successfully installed, showing up to the room on time, having a robot that conforms to the design requirements, and having your robot attempt the course.
- **Detect the wall**: 10 points. You will receive these points if your robot stops or attempts to turn right when it approaches the wall.
- **Turn at the wall**: 10 points. You will receive these points if your robot successfully turns right and starts following the wall.
- **Following the wall**: 20 points. You will receive 20 points if your robot is never more than the allowable maximum distance from the wall. This requirement must be satisfied 40 cm after the robot is supposed to turn right until the destination. Partial credit will be awarded.
- **Accuracy of reaching goal**: 20 points. Points will be based on the accuracy of reaching the robot's target location when it stops at the end of the task. Full points will be awarded if the robot stops inside the goal zone ± 5cm, where it ideally should end. Partial credit will be given based on the distance to this goal zone.

Penalties:

- There will be a 10-point penalty if, after crossing the start line, your robot ever exits the 3m by 3m workspace.
- There will be a 10-point penalty if more than 90 seconds elapse from when the dark gray center button is pushed at the start location until the robot stops.
- If the robot is still moving after 2 minutes, then the instructor can stop the run, and no additional points can be earned.

Code

To complete this lab, you will use a subset of the EV3 MicroPython API:
https://pybricks.com/ev3-micropython/index.html
Important: You may only use the following modules from EV3 MicroPython API (covered in chapters 3-6 of the API document above):
ev3brick
ev3devices
parameters
tools
You may ***not*** import, use, or copy the "**robotics**" module, or any of its included classes or functions, in your program (covered in chapter 7 of the API document above).

A robot relying on a module not allowed as stated above will be disqualified. Also, any robot using Wi-Fi or Bluetooth communications in any way will be disqualified.

You must submit to **Gradescope** a zip file of your Python source code files. The zip file should include all the Python source code files you wrote (which should have a .py file extension). Please include the **names** and **PIDs** of all team members, along with the **team number,** in the comments at the top of all included Python source files.

Optional: Lab Reflections

Optional reflections will award points towards the performance portion of the lab and are designed to prompt students to think systematically and critically about the tasks. You must have attempted a lab in order to receive credit for a post-lab reflection. These should be submitted as a *PDF* on **Gradescope** for each team. Each reflection you complete is worth 20 points and will scale the performance part of your lab score (i.e., the 90 points excluding turning in your code) to be out of 70 or 50 points. For example, if you do both reflections, turn in your code on time, and get 70/90 for robot performance, your score will be 10 (for turning in code) + 40 (for doing both reflections) + 50*(70/90) = 89 points.

Pre-Lab Reflection

Task 1:

Write a short description of your robot design in terms of the components used and how they help accomplish the task. Examples include motors, sensors, locomotion type (wheels, tracks, legs, etc…), or any other relevant components.

Task 2:

Draw a high-level flowchart of the robot's functionality.

Task 3:

Write the pseudocode that implements your flowchart.

Post-Lab Reflection

Task 1:

Write a short description of your robot's final design and how it differs from your initial concept. Why did you choose to make those changes?

Task 2:

Consider how your robot performed in the lab and how it was different from your expectations. Were there any particular shortcomings that you identified?

Task 3:

Explain what changes you would make to your design or implementation to achieve better results for the particular task.

Collaboration

You should feel free to discuss concepts in the course with other students in natural language (e.g., English). You should not share any programming code with others and must write all the robot's programming code yourself. If you access any sources other than the textbook or documents on Canvas, you must cite them in comments at the top of your code and send an email to the instructor with the citation. You must fully understand your code and be able to reproduce the algorithmic approach without references if asked. The Honor Code is in effect for these policies.

Good luck!